

The origin of ferromagnetism and the Hubbard model

part 5: flat-band ferromagnetism

***Advanced Topics in
Statistical Physics
by Hal Tasaki***

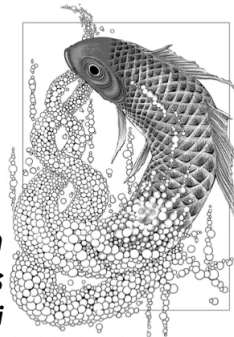


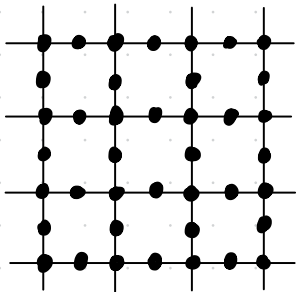
Illustration © Mari Okazaki 2020. All Rights Reserved

Stoner criterion $UD_F \gtrsim 1$

Nagaoka ferromagnetism $U = \infty$ / flat-band models $D_F = \infty$

§ Lieb's ferrimagnetism $\rightarrow$ Lieb lattice

$L \times L$ decorated square lattice Λ



$$|\Lambda| = 3L^2$$

$$t_{xy} = \begin{cases} t \in \mathbb{R} \setminus \{0\}, & x \text{ and } y \text{ are neighbors} \\ 0, & \text{otherwise} \end{cases}$$



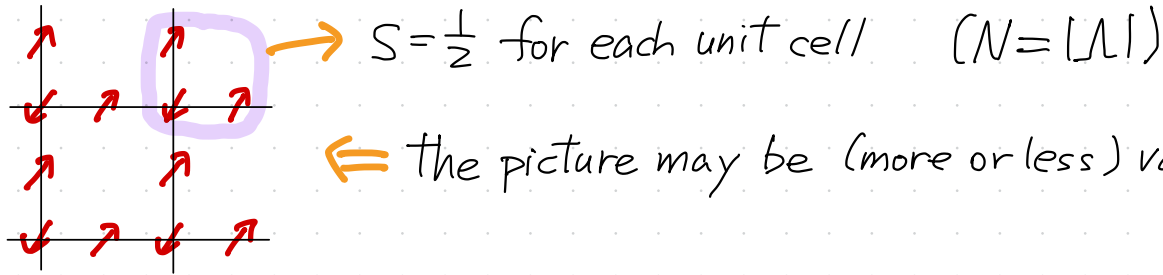
H.T. and Elliott Lieb (1932-)
Harvard 2022

Lieb 1989

theorem consider the Hubbard model with above t_{xy} and $N = |\Lambda| = 3L^2$
for any $U > 0$ the ground state has $S_{\text{tot}} = L^2/2$ and is unique
apart from the trivial $2S_{\text{tot}} + 1$ fold degeneracy

rigorously establishes the emergence of a bulk magnetization in a simple model!
only a small corollary of a beautiful (and important) general theorem $\rightarrow$ my book

► $S_{\text{tot}} = L^2/2$ suggests ferrimagnetism



⇐ the picture may be (more or less) valid for $U \gg |t|$.

► flat band

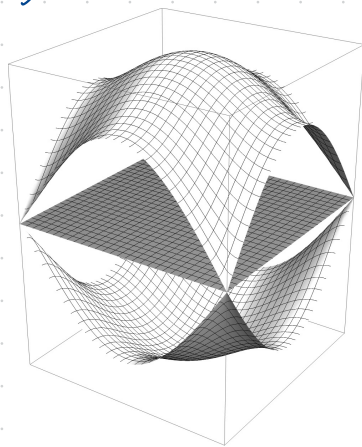
Corresponding single-particle tight-binding Schrödinger equation

$$\sum_{y \in \Lambda} t_{xy} \psi_y^{(j)} = \epsilon_j \psi_x^{(j)} \quad (x \in \Lambda) \quad (1) \quad \epsilon_j \leq \epsilon_{j+1} \quad (2)$$

flat (dispersionless) band

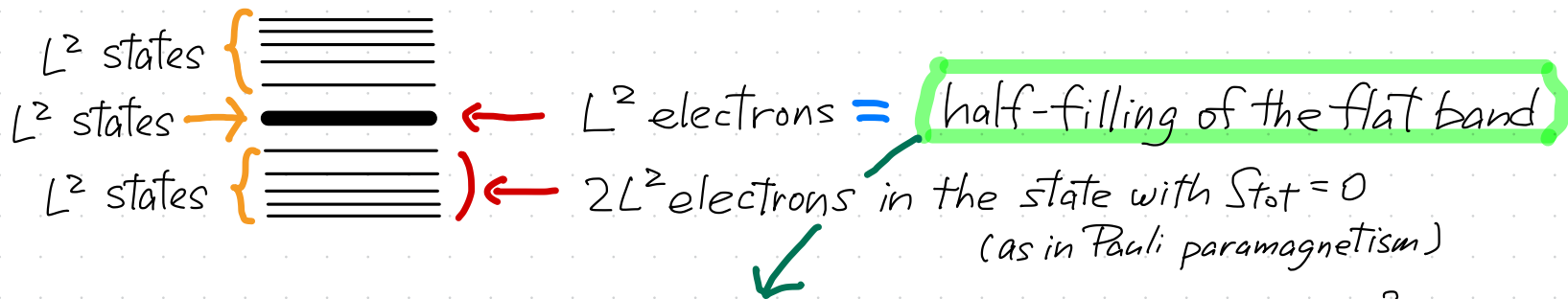
$$\epsilon_j \begin{cases} < 0, & j=1, \dots, L^2 \\ = 0, & j=L^2+1, \dots, 2L^2 \\ > 0, & j=2L^2+1, \dots, 3L^2 \end{cases} \quad (3)$$

for odd L



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ground states for $U=0$ and $N=3L^2$



highly degenerate ground states with all possible $S_{\text{tot}} \leq \frac{L^2}{2}$

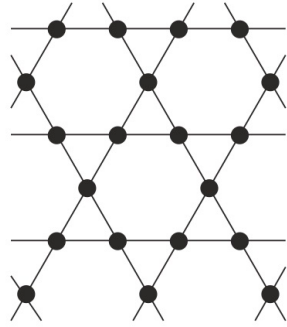
$$\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow \quad \text{---} \quad \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow \uparrow$$

$$S_{\text{tot}} = \frac{1}{2} \qquad S_{\text{tot}} = \frac{L^2}{2}$$

for any $U > 0$, the degeneracy is lifted and one only has an essentially unique ground state with $S_{\text{tot}} = \frac{L^2}{2}$

highly nontrivial!

§ Mielke's flat-band ferromagnetism



Λ kagomé lattice

$$t_{xy} = \begin{cases} t > 0, & x \text{ and } y \text{ are neighbors} \\ 0, & \text{otherwise} \end{cases}$$



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kagomé = 籠目
(mesh of woven bamboo)

Mielke 1991

theorem consider the Hubbard model with above t_{xy} and $N = \frac{|\Lambda|}{3} + 1$
for any $U > 0$, the model exhibits ferromagnetism and the ground state is unique

part 3 - p.11

an elegant example of Mielke's general theorem
for the Hubbard model on line graphs

flat band

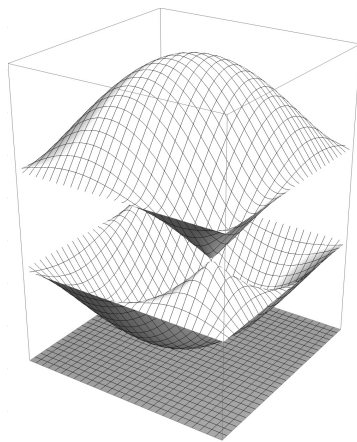
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corresponding single-particle tight-binding Schrödinger equation

$$\sum_{y \in \Lambda} t_{xy} \psi_y^{(j)} = \epsilon_j \psi_x^{(j)} \quad (x \in \Lambda) \quad (1) \quad \epsilon_j \leq \epsilon_{j+1} \quad (2)$$

flat (dispersionless) band

$$\epsilon_j \begin{cases} = -2t, & j=1, 2, \dots, \frac{|\Lambda|}{3} + 1 \\ > -2t, & j = \frac{|\Lambda|}{3} + 2, \dots, |\Lambda| \end{cases} \quad (3)$$



for $U=0$, ground states are linear combinations of

$$\left(\prod_{j \in S_{\uparrow}} \hat{a}_{j\uparrow}^{\dagger} \right) \left(\prod_{j \in S_{\downarrow}} \hat{a}_{j\downarrow}^{\dagger} \right) |\Phi_{\text{vac}}\rangle \quad (4)$$

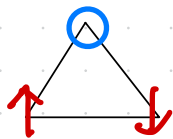
$$\text{with } S_{\uparrow}, S_{\downarrow} \subset \{1, 2, \dots, \frac{|\Lambda|}{3} + 1\}, \quad |S_{\uparrow}| + |S_{\downarrow}| = N \quad (5)$$

highly degenerate with all possible $S_{\text{tot}} \leq N/2$

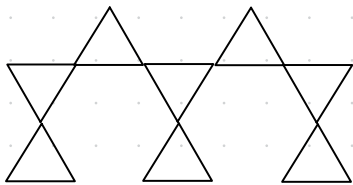
for $U > 0$ the degeneracy is lifted $\rightarrow$ ferromagnetism!

see
part 3 - p. 14
for the
simplest
example

§ intermission — H.T. after the Nagaoka ferromagnetism paper



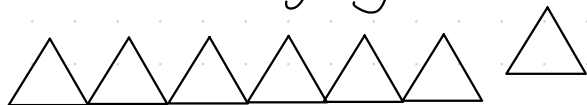
the triangle model should be a key for ferromagnetism!
combine multiple triangles to get bulk ferromagnetism



▶ noticed that kagomé at $1/6$ filling
has ferromagnetic ground states

also (correctly) realized that the uniqueness is hard to prove

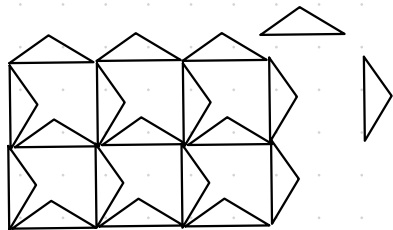
▶ started designing artificial models that were easier to treat



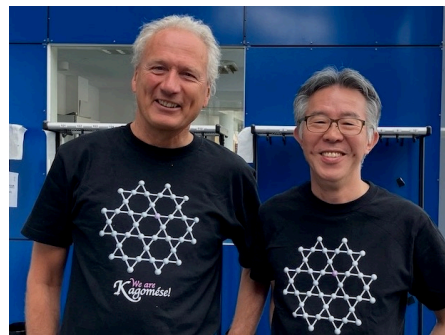
Tasaki, PRL 1992

realized Mielke's works after writing this up's

Mielke - Tasaki, CMP 1993

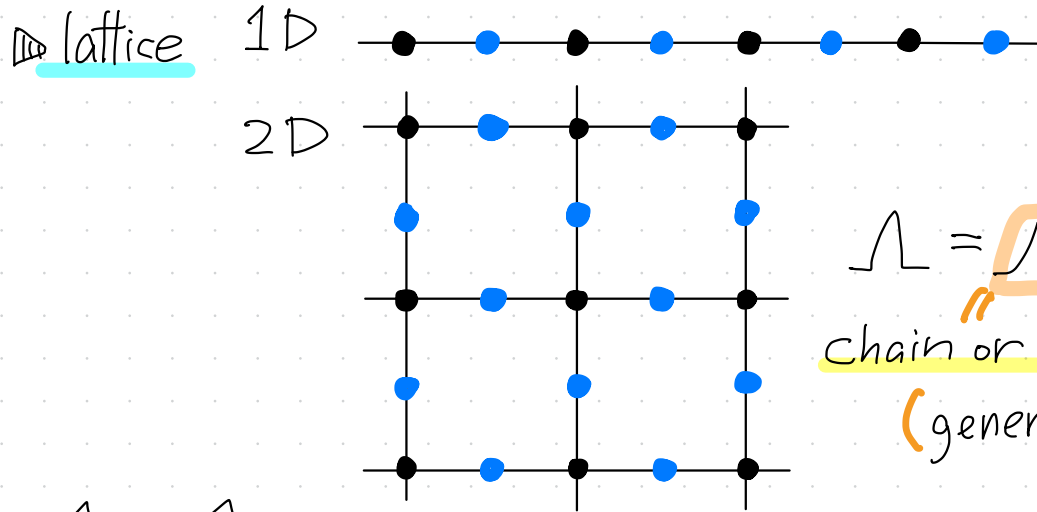


Andreas Mielke and Hal Tasaki
 Munich, 2025



§ Tasaki's flat-band ferromagnetism

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$$\Lambda = \mathcal{M} \cup \mathcal{O} \quad (1)$$

chain or square lattice with spacing 1
(generalization to $d \geq 3$ is trivial)

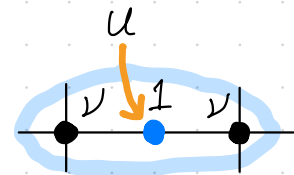
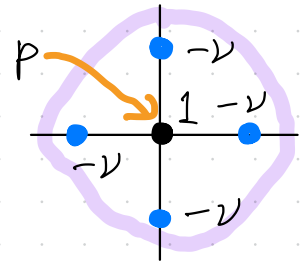
$\hat{a}$ and $\hat{b}$ operators $\nu > 0$ model parameter

$$p \in \mathcal{M}, \quad \hat{a}_{p\sigma} = \hat{c}_{p\sigma} - \nu \sum_{u \in \mathcal{O}} \hat{c}_{u\sigma} \quad (2)$$

$|u - p| = 1/2$

$$u \in \mathcal{O}, \quad \hat{b}_{u\sigma} = \hat{c}_{u\sigma} + \nu \sum_{p \in \mathcal{M}} \hat{c}_{p\sigma} \quad (3)$$

$|p - u| = 1/2$

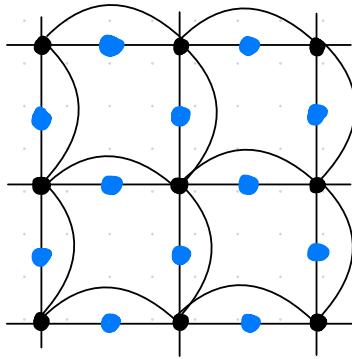


▶ Hamiltonian and the main theorem

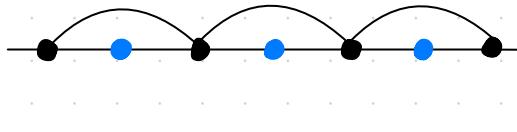
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$$t > 0, U > 0$$

$$\hat{H} = t \sum_{\substack{u \in \mathcal{O} \\ \sigma = \uparrow, \downarrow}} \hat{b}_{u\sigma}^\dagger \hat{b}_{u\sigma} + U \sum_{x \in \mathcal{L}} \hat{n}_{x\uparrow} \hat{n}_{x\downarrow} \quad (1)$$



written as $\sum_{\substack{x, y \in \mathcal{L} \\ \sigma = \uparrow, \downarrow}} t_{xy} \hat{c}_{x\sigma}^\dagger \hat{c}_{y\sigma}$
with suitable t_{xy}



theorem consider the above Hubbard model with $N = |\mathcal{M}|$ Tasaki 1992
for any $U > 0$, the model exhibits ferromagnetism and the ground state is unique

part 3 - p. 11

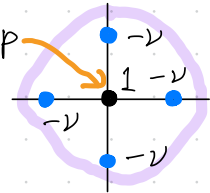
flat band

corresponding single-particle tight-binding Schrödinger equation

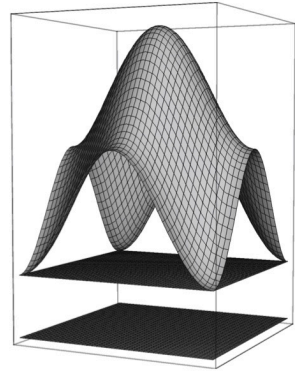
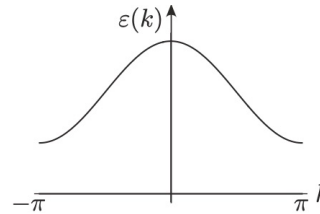
$$\sum_{y \in \Lambda} t_{xy} \psi_y^{(j)} = \epsilon_j \psi_x^{(j)} \quad (x \in \Lambda) \quad (1) \quad \epsilon_j \leq \epsilon_{j+1} \quad (2)$$

$$\epsilon_j \begin{cases} = 0, & j=1, 2, \dots, |\mathcal{M}| \\ > 0, & j=|\mathcal{M}|+1, \dots, |\mathcal{N}| \end{cases} \quad \leftarrow \text{flat (dispersionless) band} \quad (3)$$

the flat band is spanned
by the $\hat{a}$ -states



unlike in Lieb's and Mielke's models,
 t_{xy} is fine-tuned to have a flat band



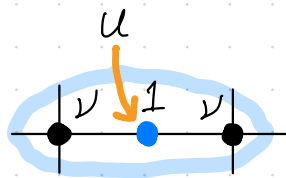
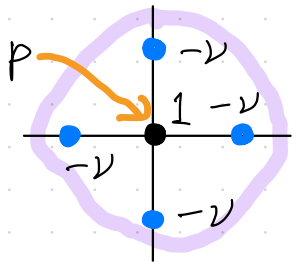
the picture of flat-band ferromagnetism for $U > 0$ is
similar to that of Mielke's ferromagnetism

BUT the proof is much easier!

proof (Tasaki's flat-band ferromagnetism)

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$$p \in \mathcal{M}, \hat{a}_{p\sigma} = \hat{c}_{p\sigma} - \nu \sum_{\substack{u \in \mathcal{O} \\ |u-p|=1/2}} \hat{c}_{u\sigma} \quad (1) \quad u \in \mathcal{O}, \hat{b}_{u\sigma} = \hat{c}_{u\sigma} + \nu \sum_{\substack{p \in \mathcal{M} \\ |p-u|=1/2}} \hat{c}_{p\sigma} \quad (2)$$



these operators span the whole space of single-electron operators

$$\hat{c}_{x\sigma} = \sum_{p \in \mathcal{M}} \alpha_p^{(x)} \hat{a}_{p\sigma} + \sum_{u \in \mathcal{O}} \beta_u^{(x)} \hat{b}_{u\sigma} \quad (3) \text{ for some } \alpha_p^{(x)}, \beta_u^{(x)} \in \mathbb{R}$$

(one can express anything by $\hat{a}$ and $\hat{b}$)

$\hat{a}$ -states and $\hat{b}$ -states are orthogonal

$$\{\hat{a}_{p\sigma}^\dagger, \hat{b}_{u\tau}\} = 0 \quad (4) \text{ for any } p, u, \sigma, \tau$$

▷ some useful facts about nonnegative operators

(finite dimensional Hilbert space)

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$$\hat{A} \geq 0 \stackrel{\text{def}}{\iff} \langle \Phi | \hat{A} | \Phi \rangle \geq 0 \text{ for } \forall |\Phi\rangle \stackrel{\text{th.}}{\iff} \hat{A}^\dagger = \hat{A} \text{ and alleigenvalues of } \hat{A} \geq 0$$

th. for $\hat{A} \geq 0$, $\hat{A}|\Phi\rangle = 0 \iff \langle \Phi | \hat{A} | \Phi \rangle = 0$ (proof: spectral decomposition)

$$\text{th. } \hat{A}_1, \dots, \hat{A}_n \geq 0 \Rightarrow \sum_{j=1}^n \hat{A}_j \geq 0$$

$$\text{th. } \hat{A}_1, \dots, \hat{A}_n \geq 0 \text{ and } \left(\sum_{j=1}^n \hat{A}_j \right) |\Phi\rangle = 0 \Rightarrow \hat{A}_j |\Phi\rangle = 0 \text{ for each } j$$

$$\text{(proof } \sum_j \langle \Phi | \hat{A}_j | \Phi \rangle = 0 \Rightarrow \langle \Phi | \hat{A}_j | \Phi \rangle = 0 \text{ for each } j)$$

$$\text{th. for any } \hat{B}, \hat{B}^\dagger \hat{B} \geq 0 \text{ (proof } \langle \Phi | \hat{B}^\dagger \hat{B} | \Phi \rangle = \|\hat{B}|\Phi\rangle\|^2)$$

$$\text{th. for any } \hat{B}_1, \dots, \hat{B}_n, \left(\sum_{j=1}^n \hat{B}_j^\dagger \hat{B}_j \right) |\Phi\rangle = 0 \Rightarrow \hat{B}_j |\Phi\rangle = 0 \text{ for each } j$$

$$\hat{H} = t \sum_{u \in \mathcal{O}} \hat{b}_{u\sigma}^\dagger \hat{b}_{u\sigma} + U \sum_{x \in \Lambda} \hat{n}_{x\uparrow} \hat{n}_{x\downarrow} \quad (1) \quad t > 0, U > 0$$

$\hat{b}_{u\sigma}^\dagger \hat{b}_{u\sigma} \geq 0$ $(\hat{c}_{x\uparrow} \hat{c}_{x\downarrow})^\dagger (\hat{c}_{x\uparrow} \hat{c}_{x\downarrow}) \geq 0$

$$\hat{H} \geq 0 \Rightarrow E_{GS} \geq 0 \quad (2)$$

ferromagnetic state $|\Phi_\uparrow\rangle = \left(\prod_{p \in \mathcal{M}} \hat{a}_{p\uparrow}^\dagger \right) |\Phi_{vac}\rangle \quad (3)$

$$p \in \mathcal{O} \Rightarrow \hat{b}_{u\sigma}^\dagger \hat{b}_{u\sigma} |\Phi_\uparrow\rangle = 0 \quad (4) \quad \text{and clearly } \hat{n}_{x\uparrow} \hat{n}_{x\downarrow} |\Phi_\uparrow\rangle = 0 \quad (5)$$

$$\hat{H} |\Phi_\uparrow\rangle = 0 \quad (6) \Rightarrow E_{GS} = 0 \quad \text{and } |\Phi_\uparrow\rangle \text{ is a ground state!}$$

$$|\Phi_M\rangle = (\hat{S}_{\text{tot}}^-)^{\frac{N}{2} - M} |\Phi_\uparrow\rangle \quad (6) \quad \text{is also a ground state}$$

$$M = -\frac{N}{2}, -\frac{N}{2} + 1, \dots, \frac{N}{2} \quad (7)$$

$$[\hat{S}_{\text{tot}}^-, \hat{H}] = 0$$

are these $N+1$ states the only ground states?

arbitrary ground state $|\Phi\rangle$

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$$|\Phi\rangle = \sum_{\substack{A_\uparrow, A_\downarrow \subset \mathcal{M} \\ B_\uparrow, B_\downarrow \subset \mathcal{O} \\ (|A_\uparrow| + |A_\downarrow| + |B_\uparrow| + |B_\downarrow| = N)}} f(A_\uparrow, A_\downarrow, B_\uparrow, B_\downarrow) (\prod_{P \in A_\uparrow} \hat{a}_{P\uparrow}^\dagger) (\prod_{P \in A_\downarrow} \hat{a}_{P\downarrow}^\dagger) (\prod_{U \in B_\uparrow} \hat{b}_{U\uparrow}^\dagger) (\prod_{U \in B_\downarrow} \hat{b}_{U\downarrow}^\dagger) |\Phi_{vac}\rangle \quad (1)$$

any N electron state can be expanded like this

$$\hat{H}|\Phi\rangle = 0 \quad (2) \Rightarrow \begin{cases} \hat{b}_{u\sigma}|\Phi\rangle = 0 \text{ for } \forall u \in \mathcal{O} \text{ and } \sigma = \uparrow, \downarrow & (3) \\ \hat{c}_{x\downarrow}\hat{c}_{x\uparrow}|\Phi\rangle = 0 \text{ for } \forall x \in \mathcal{L} = \mathcal{M} \cup \mathcal{O} & (4) \end{cases}$$

P12-(1), P11

step 1 restriction to the lowest $\hat{a}$ -band

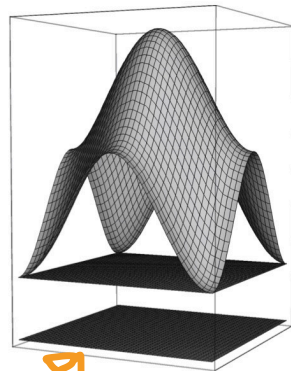
(3) $\Rightarrow$ there can be no $\hat{b}^\dagger$ in $|\Phi\rangle$

$$f(A_\uparrow, A_\downarrow, B_\uparrow, B_\downarrow) = 0 \text{ unless } B_\uparrow = B_\downarrow = \emptyset \quad (5)$$

we thus have

$$|\Phi\rangle = \sum_{\substack{A_\uparrow, A_\downarrow \subset \mathcal{M} \\ (|A_\uparrow| + |A_\downarrow| = N)}} g(A_\uparrow, A_\downarrow) (\prod_{P \in A_\uparrow} \hat{a}_{P\uparrow}^\dagger) (\prod_{P \in A_\downarrow} \hat{a}_{P\downarrow}^\dagger) |\Phi_{vac}\rangle \quad (6)$$

$\hat{a}$ -band



$$|\Phi\rangle = \sum_{\substack{A_\uparrow, A_\downarrow \subset \mathcal{M} \\ (|A_\uparrow| + |A_\downarrow| = N)}} g(A_\uparrow, A_\downarrow) \left(\prod_{p \in A_\uparrow} \hat{a}_{p\uparrow}^\dagger \right) \left(\prod_{p \in A_\downarrow} \hat{a}_{p\downarrow}^\dagger \right) |\Phi_{\text{vac}}\rangle \quad (1)$$

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step 2 reduction to a "spin system"

$$p(3)-(4) \Rightarrow \hat{c}_{p\downarrow} \hat{c}_{p\uparrow} |\Phi\rangle = 0 \text{ for any } p \in \mathcal{M} \quad (2)$$

$$\text{note that } \hat{c}_{p\downarrow} \hat{c}_{p\uparrow} \hat{a}_{p\uparrow}^\dagger \hat{a}_{p\downarrow}^\dagger |\Phi_{\text{vac}}\rangle = |\Phi_{\text{vac}}\rangle \quad (3)$$

$\swarrow \hat{c}_{p\uparrow}^\dagger \searrow \hat{c}_{p\downarrow}^\dagger$

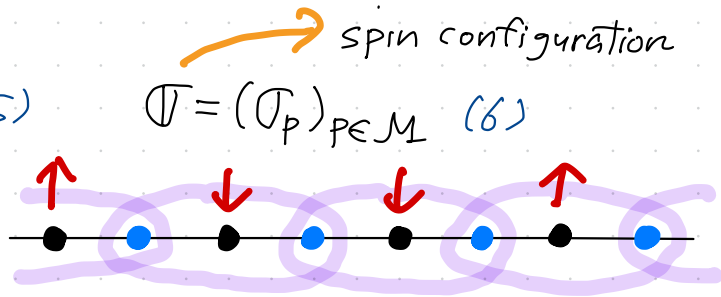
$$\text{thus } g(A_\uparrow, A_\downarrow) = 0 \text{ if } A_\uparrow \cap A_\downarrow \neq \emptyset \quad (4)$$

$$A_\uparrow \cap A_\downarrow = \emptyset \text{ and } |A_\uparrow| + |A_\downarrow| = N = |\mathcal{M}| \implies A_\uparrow \cup A_\downarrow = \mathcal{M}$$

every metal site $p \in \mathcal{M}$ is occupied by an electron in the α -state

$$|\Phi\rangle = \sum_{\mathbb{D}} h(\mathbb{D}) \left(\prod_{p \in \mathcal{M}} \hat{a}_{p\sigma_p}^\dagger \right) |\Phi_{\text{vac}}\rangle \quad (5)$$

$\mathbb{D} = (\sigma_p)_{p \in \mathcal{M}} \quad (6)$



Step 3 exchange interaction

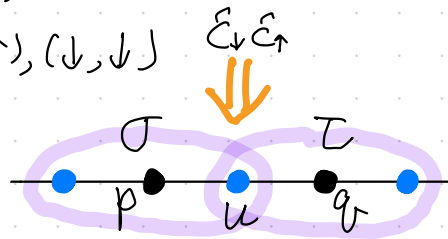
$$p|3-(4) \Rightarrow \hat{C}_{u\downarrow} \hat{C}_{u\uparrow} |\Phi\rangle = 0 \text{ for any } u \in \mathcal{O} \quad (1)$$

Take $p, q \in \mathcal{M}$ with $|p-q|=1$ and let $u \in \mathcal{O}$ be the site between p and q

$$\text{Then } \hat{C}_{u\downarrow} \hat{C}_{u\uparrow} \hat{a}_{p\sigma}^\dagger \hat{a}_{q\tau}^\dagger |\Phi_{vac}\rangle = \begin{cases} \nu^2 |\Phi_{vac}\rangle, & (\sigma, \tau) = (\uparrow, \downarrow) \\ -\nu^2 |\Phi_{vac}\rangle, & (\sigma, \tau) = (\downarrow, \uparrow) \\ 0, & (\sigma, \tau) = (\uparrow, \uparrow), (\downarrow, \downarrow) \end{cases} \quad (2)$$

$\underbrace{\hat{C}_{u\downarrow} \hat{C}_{u\uparrow}}_{-\nu \hat{C}_{u\sigma}^\dagger} \hat{a}_{p\sigma}^\dagger \hat{a}_{q\tau}^\dagger \xrightarrow{-\nu \hat{C}_{u\tau}^\dagger}$

$$\text{write } \mathcal{M} = \{1, 2, 3, \dots, N\} \quad (3)$$



$$p|4-(5) \quad |\Phi\rangle = \sum_{\sigma_1, \sigma_2, \sigma_3, \dots, \sigma_N} h(\sigma_1, \sigma_2, \sigma_3, \dots, \sigma_N) \hat{a}_{1\sigma_1}^\dagger \hat{a}_{2\sigma_2}^\dagger \left(\prod_{p=3}^N \hat{a}_{p\sigma_p}^\dagger \right) |\Phi_{vac}\rangle \quad (4)$$

$u = 3/2$ be the site between 1 and 2. then (2) implies

$$\hat{C}_{u\downarrow} \hat{C}_{u\uparrow} |\Phi\rangle = \nu^2 \sum_{\sigma_3, \dots, \sigma_N} \{ h(\uparrow, \downarrow, \sigma_3, \dots, \sigma_N) - h(\downarrow, \uparrow, \sigma_3, \dots, \sigma_N) \} \left(\prod_{p=3}^N \hat{a}_{p\sigma_p}^\dagger \right) |\Phi_{vac}\rangle \quad (5)$$

$$(1) \Rightarrow h(\uparrow, \downarrow, \sigma_3, \dots, \sigma_N) = h(\downarrow, \uparrow, \sigma_3, \dots, \sigma_N) \text{ for any } \sigma_3, \dots, \sigma_N \quad (6)$$

for $p, q \in \mathcal{M}$ s.t. $|p - q| = 1$, $h(\sigma) = h(\sigma_{p \leftrightarrow q})$ (1)

thus spin configuration obtained by swapping σ_p and σ_q

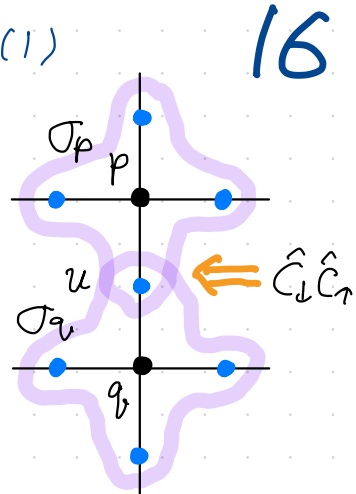
$$h(\sigma) = h(\tau) \text{ if } \sum_{p \in \mathcal{M}} \sigma_p = \sum_{p \in \mathcal{M}} \tau_p \quad (2)$$

$$p(4-5) \quad |\Phi\rangle = \sum_{\sigma} h(\sigma) \left(\prod_{p \in \mathcal{M}} \hat{a}_{p\sigma_p}^\dagger \right) |\Phi_{vac}\rangle \quad (3)$$

$|\Phi\rangle$ is a linear combination of

$$|\Phi_M\rangle = (\hat{S}_{tot}^-)^{\frac{N}{2} - M} \left(\prod_{p \in \mathcal{M}} \hat{a}_{p\uparrow}^\dagger \right) |\Phi_{vac}\rangle \quad (4)$$

$$\text{with } M = -\frac{N}{2}, -\frac{N}{2} + 1, \dots, \frac{N}{2} \quad (5)$$

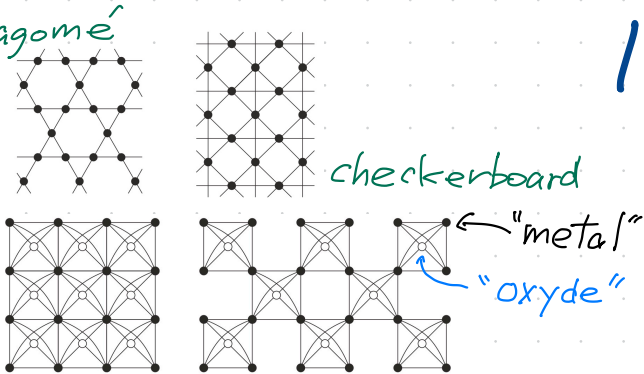


§ flat-band ferromagnetism ^{early 1990s} ^{kagomé}

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Mielke model line graph 

Tasaki model cell construction 



- specifically designed tight-binding models with highly degenerate single-particle ground states
- for the electron number N equal to the degree of the degeneracy, many-body ground states for $U=0$ are also highly degenerate and have any $S_{\text{tot}} \leq \frac{N}{2}$
- Coulomb interaction $U > 0$ lifts the degeneracy and one has saturated ferrimagnetism

(remark a flat-band does NOT always lead to ferromagnetism)
trivial example $t_{xy}=0$ for all x, y

- a ground state of $\hat{H}$ is a simultaneous ground state of $\hat{H}_{\text{hop}}$ and $\hat{H}_{\text{int}}$
NO "competition" (or frustration) between $\hat{H}_{\text{hop}}$ and $\hat{H}_{\text{int}}$  part 3 - p10

§ beyond flat-band ferromagnetism

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is flat-band ferromagnetism stable against modification of the hopping t_{xy} ?

↓ various studies in 1990s

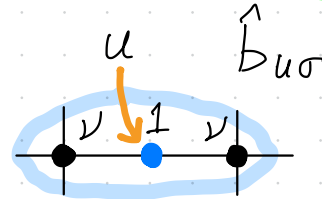
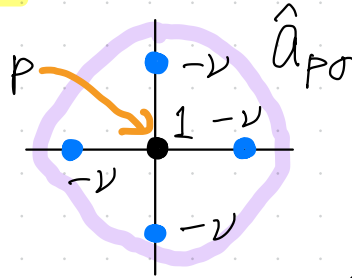
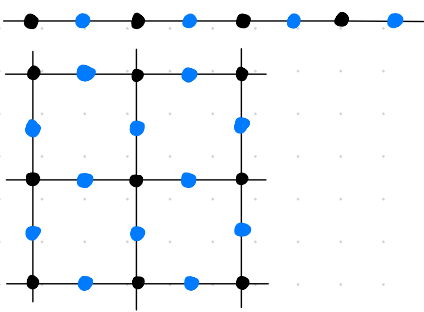
yes! provided that $U > 0$ is sufficiently large

the first rigorous example Tasaki 1995

"healthy" insulating ferromagnet
without singular limits

$$D_F < \infty, U < \infty$$

the same lattice $\Lambda = \mathbb{M} \cup \Theta$



the same operators $\hat{a}_{p\sigma}, \hat{b}_{u\sigma}$

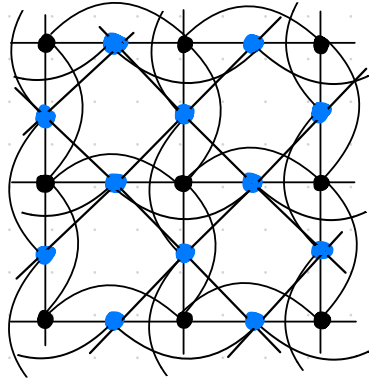
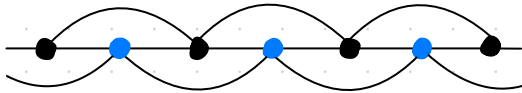
$$\hat{H} = t \sum_{u \in \Theta} \sum_{\sigma = \uparrow, \downarrow} \hat{b}_{u\sigma}^\dagger \hat{b}_{u\sigma} - s \sum_{p \in \mathbb{M}} \sum_{\sigma = \uparrow, \downarrow} \hat{a}_{p\sigma}^\dagger \hat{a}_{p\sigma} + U \sum_{x \in \Lambda} \hat{n}_{x\uparrow} \hat{n}_{x\downarrow} \quad (1)$$

$$t, s, U > 0$$

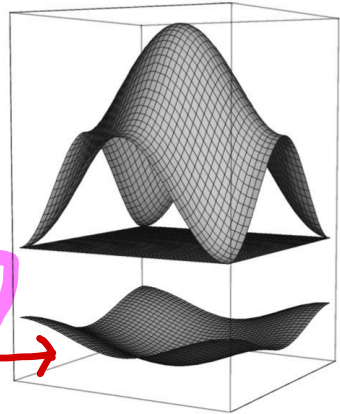
$$\hat{H} = t \sum_{u \in \mathcal{O}} \sum_{\sigma=\uparrow, \downarrow} \hat{b}_{u\sigma}^\dagger \hat{b}_{u\sigma} - s \sum_{p \in \mathcal{M}} \sum_{\sigma=\uparrow, \downarrow} \hat{a}_{p\sigma}^\dagger \hat{a}_{p\sigma} + U \sum_{x \in \mathcal{L}} \hat{n}_{x\uparrow} \hat{n}_{x\downarrow} \quad (1) \quad t, s, U > 0 \quad 19$$

$$\sum_{x,y,\sigma} t_{xy} \hat{c}_{x\sigma}^\dagger \hat{c}_{y\sigma}$$

Somewhat complicated but still short-ranged hopping t_{xy}



NO LONGER FLAT!!



Pauli paramagnetism for $U=0$ (and probably for small $U>0$)
competition (frustration) between $\hat{H}_{\text{hop}}$ and $\hat{H}_{\text{int}}$!!

Tasaki 1995, 2003

theorem consider the above Hubbard model with $N=|\mathcal{M}|$ and $U \leq U_c(d)$
 for sufficiently large t/s and U/s , the model exhibits ferromagnetism
and the ground state is unique  part 3 - p.11 

we also have a rigorous estimate of the spin-wave excitation energy Tasaki 1994

idea of the proof

$$\hat{H} = \text{constant} + \sum_{p \in \mathcal{M}} \hat{h}_p \quad (1)$$

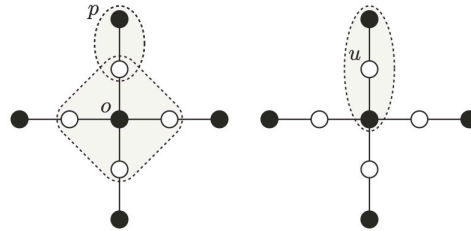
the ground state is a simultaneous ground state of all $\hat{h}_p$ with $p \in \mathcal{M}$

20

430

11 The Origin of Ferromagnetism

Fig. 11.23 The nine-site lattice on which the local Hamiltonian $\hat{h}_o$ for $d = 2$ acts. We have also depicted the states generated by $\hat{a}_o^\dagger$ and $\hat{a}_p^\dagger$ (left), and by $\hat{b}_u^\dagger$ (right) (© Hal Tasaki 2020. All Rights Reserved)



$$\begin{aligned} \hat{h}_p = & (1 + 2dv^2)s - s \sum_{\sigma=\uparrow,\downarrow} \hat{a}_{p,\sigma}^\dagger \hat{a}_{p,\sigma} + \frac{t-\lambda}{2} \sum_{\substack{u \in \mathcal{J} \\ (|u-p|=1/2) \\ \sigma=\uparrow,\downarrow}} \hat{b}_{u,\sigma}^\dagger \hat{b}_{u,\sigma} \\ & + (1-\kappa)(U-\lambda)\hat{n}_{p,\uparrow}\hat{n}_{p,\downarrow} + \frac{U-\lambda}{2} \sum_{\substack{u \in \mathcal{J} \\ (|u-p|=1/2)}} \hat{n}_{u,\uparrow}\hat{n}_{u,\downarrow} \\ & + \frac{\kappa}{2d}(U-\lambda) \sum_{\substack{q \in \mathcal{S} \\ (|q-p|=1)}} \hat{n}_{q,\uparrow}\hat{n}_{q,\downarrow}. \end{aligned} \quad (11.4.48)$$

